Carl Zeiss Vision International GmbH
73430 Aalen

On the left in the address field, press the **ENTER key** to start a new line.

● Please use the **TAB key** or **the mouse key** to jump from field to field (do **not** use the ENTER key ●

Carl Zeiss Vision International GmbH

Turnstrasse 27
73430 Aalen
Germany

Phone: +49 (0)73 61 591-0

Fax: +49 (0)73 61 591-1312

Email: wendelin.niesl@zeiss.com

Division/Dept.: Business Enablement & IT

Date 10.06.2026

.NET & Angular Development Hands-On Assessment

We are pleased to present a coding challenge tailored for a Software Engineer with experience in .NET and frontend development using Angular. This assessment assumes familiarity with ASP.NET Core, REST APIs, and modern frontend frameworks.

Submission Guidelines:
You are required to submit your project or a link to your work within 7 days of receiving this assessment to proceed with the next steps in our evaluation process.

Development Standards:
You are encouraged to utilize any design patterns and error handling methodologies that align with industry best practices. Please ensure that your implementation adheres to coding standards typically employed in professional projects. You are welcome to enhance the specified tasks with additional features that you believe would improve the overall quality of the project. Should you have any questions, please do not hesitate to reach out via email.

Project Overview:
Your task is to develop an API that manages products, exposing RESTful endpoints that provide standard CRUD functionality for the Product entity.

In addition, you are expected to create a frontend application using Angular that consumes the provided API and presents the data in a meaningful and user-friendly way.

CRUD Endpoints

- **GET** /api/products
- **POST** /api/products
- **GET** /api/products/{id}
- **PUT** /api/products/{id}
- **DELETE** /api/products/{id}

Additional Endpoints

- **POST** /api/products/{id}/decrement-stock/{quantity}
  This endpoint should return an OK response after decrementing the stock by the specified quantity.
- **POST** /api/products/{id}/add-to-stock/{quantity}
  This endpoint should return an OK response after incrementing the stock by the specified quantity.
- **GET** /api/products/search?name={name}
  This endpoint should allow searching for products by name, supporting partial matches.
- **GET** /api/products/stock-level?min={min}&max={max}
  This endpoint should return products within a specified stock range.

Requirements:
- The product identifier should be auto-generated as a unique 6-digit number. Even when using multiple instances at the same time, no duplicate IDs should be generated.
- Implement validation for product creation and updates, ensuring that all required fields are present and have valid values.

- Every product must possess a unique identifier. When returning a list of products from any of the aforementioned endpoints, the available stock for each product must be included in the response.

Database Requirements:
Please create a database using Entity Framework Core (EFCore) migrations with a code-first approach. The database schema supports the additional fields and relationships you define.Implement database seeding to populate the database with initial data for testing.

Frontend Extension

Objective:
Create a frontend application using Angular that consumes the provided API and presents the data in a meaningful and user-friendly way.

Requirements:
- Integrate with the API developed in this assignment
- Consume the available endpoints and display the data appropriately
- Provide a user interface that allows interaction with the product data
- Ensure that the application reflects changes made through the API (e.g. updates to product data or stock levels)

Technical Expectations:
- Use Angular (latest stable version recommended)
- Structure the application in a clean, maintainable way
- Follow common best practices for Angular development
- Handle asynchronous data and API communication properly

Notes:
- The exact design and structure of the UI are intentionally left open
- Emphasis is placed on how data is consumed, structured, and presented
- Candidates are encouraged to make reasonable assumptions and decisions

Optional:
- Improve usability and user experience
- Add validation, filtering, or interaction patterns where appropriate
- Apply styling or UI frameworks of choice

Self-Check List Before Submission

☐ Ensure that your code is clean, well-formatted, executable, and testable.
☐ Apply best practices and design patterns where appropriate.
☐ Confirm that all 9 required endpoints are implemented.
☐ Include logic for product ID generation suitable for a distributed environment with multiple instances.
☐ Utilize EF migrations with a code-first approach to create the database.
☐ Provide unit test cases for various scenarios.
☐ Share your code along with documentation for running the project locally.

Optional: Behavior-Driven Development (BDD) tests to validate expected user outcomes.

We look forward to reviewing your submission!

Best Regards,

Carl Zeiss Vision International
Business Enablement & IT Team